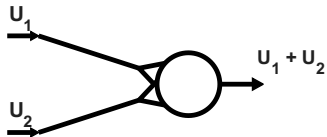
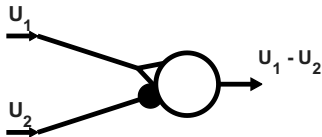


(a) addition



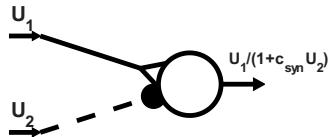
two excitatory signal-transmission synapses

(b) subtraction



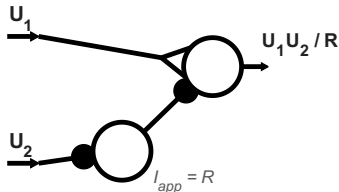
inhibitory synapse weight = $-k_{\text{syn},1} \Delta E_{s,1} / \Delta E_{s,2}$

(c) division



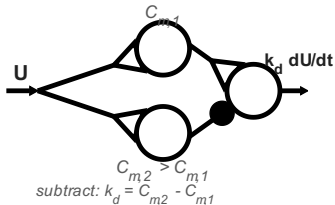
shunting: $\Delta E_{s,2} = 0$; dashed = modulatory

(d) multiplication



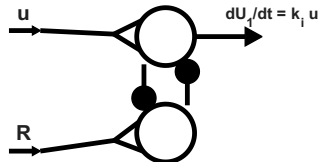
disinhibitory cascade: modulatory synapses in series

(e) differentiation



subtract: $k_d = C_{m2} - C_{m1}$

(f) integration



mutual disinhibition = line attractor (no leak)